



MYTHIC

Compiler Optimization Report - ResNet-50

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1 Summary

For ResNet-50 at 224x224x3 resolution:

Performance simulation estimates a Total Processing Time (for the compiled ONNX file) of

- 0.55 ms, of which
- 0.44 ms is Analog NPU processing time, which corresponds to
 - 71.6% ACE utilization, using 24 Analog Compute Engines; and of which
- 0.11 ms is Digital NPU processing time,
 - using only 1 Digital NPU core.

The number of used Digital NPU cores (1) was selected to show the performance of a single core. If more Digital NPU cores were used, the Digital NPU processing time could be substantially shortened. For example, 8 Digital NPU cores could reduce the Digital NPU processing time by ~8x, resulting in a total (Analog + Digital) processing time of ~0.45 ms, corresponding to ~2200 fps.

There is a slight digital bottleneck, due to SRAM Read/Write time, which increases the Analog NPU processing time by ~2%. This will be addressed in the future by improving compiler modeling/optimization of SRAM accesses. A reduction in SRAM Read/Write time by ~30% should be possible, but only ~2% is needed.

Automatic optimization performs as well as the best manual optimization.

We may be able to improve performance in the future by finding a more optimal placement of weight blocks to ACEs (to increase the ACE utilization), but note that the current ACE utilization of 71.6% is already quite high.

We may be able to improve performance in the future by allowing weight block widths to exceed 256. The hardware design recently increased support to 512, and the compiler will add support for 512 in the future.

The estimated power is

- 1.1 Watts at the maximum frame rate (1830 fps), if restricted to using only 1 Digital NPU core.

Power estimates are approximate, and will be more precise in the future.

2 Reproducing the results

The results in this report were produced using the “high_optimization” compiler config for ResNet-50.

Note that some variance in results is expected (up to a few percent), due to non-deterministic, multi-threaded compiler optimizations.

The `mythic-compiler` command was

```
mythic-compiler \  
  --input-artifact $MYTHIC_SDK_ROOT/models/training/huggingface_classifiers/  
  resnet50_imagenet_trained.tar.gz \  
  --compiler-config $MYTHIC_SDK_ROOT/mythic-model-zoo/configs/huggingface_classifiers/  
  compiler/resnet50_imagenet_224x224x3_m2024_high_optimization.yaml \  
  --output-artifact /rscratch/tonbomythic3/<your_username>/compiler/  
  resnet50_imagenet_224x224x3_m2024_high_optimization_compiled.tar.gz
```

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The `mythic-ppa-estimators` command for the maximum frame rate (1830 fps) was

```
mythic-ppa-estimators \
  --estimate-performance \
  --estimate-power \
  /rscratch/tonbomythic3/<your_username>/compiler/
resnet50_imagenet_224x224x3_m2024_high_optimization_compiled.tar.gz
```

3 Performance estimates

The `mythic-ppa-estimators` tool outputs the following performance statistics:

```
Analog NPU Total Estimated Processing Time for Compiled ONNX File: 0.44 ms
Analog NPU Estimated Frame Rate for Compiled ONNX File: 2,274.40 fps (frames per second)
Critical Path ACE Latency: 0.43 ms
Maximum SRAM Read/Write Time (over 6 ACE tiles): 0.44 ms
Maximum SIMD Operation Time (over 6 ACE tiles): 0.00 ms
Average SRAM Read/Write Time (averaged over 6 ACE tiles): 0.34 ms
Average SIMD Operation Time (averaged over 6 ACE tiles): 0.00 ms
Total Executed ACE Operations: 47,236
Total Executed ACE MACs: 5,324,275,712
Maximum Theoretical ACE Execution Time (With No Parallelization Using 1 ACE): 7.56 ms
Minimum Theoretical ACE Execution Time (With Even Parallelization Across 24 ACEs): 0.31
ms
ACE Utilization (Minimum Theoretical Time/Estimated Processing Time): 71.62%
Total SRAM Bytes Read Across Chip: 140,795,352 bytes
Total SRAM Bytes Written Across Chip: 96,647,544 bytes
Estimated Die Area to Achieve Estimated Processing Time: 126.55 mm^2
-----
Digital NPU Performance Metrics (from separate ONNX graph processing)
  Total Estimated MACs Targeting Digital NPU: 2,000,000
  Digital NPU MAC Utilization: 30.00%
  Digital NPU Cores: 1
  Digital NPU Frequency: 1,000 MHz
  Digital Estimated Frame Processing Time: 0.11 ms
  Digital Estimated Frame Rate: 9,375.06 fps (frames per second)
-----
Combined Analog + Digital NPU Latency: 0.55 ms
Combined Analog + Digital NPU Total Estimated Frame Rate: 1,830.36 fps (frames per
second)

NOTE: The information reported above may include the Analog NPU or the Analog
NPU and a digital NPU as indicated. This estimated processing time and
associated frame rate only covers the execution of the compiled ONNX model(s)
running on one or both of those compute solutions in isolation. It does not
include PCIe transfer overhead. Performance estimation for additional processing
```

steps and a combined compute hardware solution will be added in future versions.

4 Power estimates

The `mythic-ppa-estimators` tool outputs the following power estimate.

Power at the maximum frame rate (1830 fps):

```
Process Nodes:  Analog 28nm, Digital 5nm
Number of ACEs:  24
Number of Digital NPU Cores: 1
Inference Rate: 1830 frames/second (fps)

Analog NPU Estimated Power: 1.104 W for target ONNX file
    Functional Unit Power: 0.821 W
    Interconnect Power:   0.283 W

Digital NPU Estimated Power: 0.009 W for digital target ONNX file

Total Combined (Analog + Digital NPU) Power: 1.113 W

NOTE: Estimation for leakage, clock tree, chip I/O power will be added in future
versions.
```

5 Compiler reports

5.1 L0 IR stats

From near the end of mythic-compiler stdout:

```
===== L0 IR stats =====
      Tiles: 7
      Buffers: 321
      Launchers: 471
    ACE Launchers: 153
      ACE Weights: 26818816
    ACE calculations: 47236
    Max ACE calculations: 8316 on tile 3_2
=====
```

5.2 ACE utilization report

The compiler-generated `ace_utilization.txt`:

Summary:

- 72.39% ACE utilization is predicted by the compiler,
- assuming no digital bottlenecks (which can typically be eliminated).
- 47,236 ACE calculations are spread across 24 ACEs, which is
- 1,968.17 ACE calculations per ACE (on average).
- 2,719 ACE calculations are in the critical path, implying
- 0.44 ms predicted inference time (160 ns per ACE calculation).

Tile	ACEs	ACE calculations	ACE calculations Per ACE	Predicted Full Network ACE Critical Path	Predicted ACE Utilization
1_1	0	N/A	N/A	N/A	N/A
3_1	4	7,644	1,911.00	2,719	70.28%
4_1	4	7,763	1,940.75	2,719	71.38%
1_2	4	7,973	1,993.25	2,719	73.31%
2_2	4	7,994	1,998.50	2,719	73.50%
3_2	4	8,316	2,079.00	2,719	76.46%
4_2	4	7,546	1,886.50	2,719	69.38%
All	24	47,236	1,968.17	2,719	72.39%

5.3 Weight utilization report

The compiler-generated `weight_utilization.txt`:

Summary:

- 89.74% of analog weight storage capacity is used, using
- 26,818,816 weights out of 29,884,416 capacity, using
- 153 weight blocks.

Tile	Weight Blocks	Allocated Weights	Weight Capacity	Utilization
3_1	25	4,484,992	4,980,736	90.05%
4_1	26	4,584,960	4,980,736	92.05%
1_2	24	4,266,240	4,980,736	85.65%
2_2	25	4,552,448	4,980,736	91.40%
3_2	26	4,568,576	4,980,736	91.72%
4_2	27	4,361,600	4,980,736	87.57%
All	153	26,818,816	29,884,416	89.74%

5.4 SRAM utilization report

The compiler-generated `sram_utilization.txt`:

Summary:

- 1,603,181 bytes of SRAM is used, consisting of
- 995,744 bytes **for** buffers (62.11%) and
- 607,437 bytes **for** control flow (37.89%), using
- 6.37% of the simulated on-chip SRAM capacity.

Tile	Buffer Allocation (bytes)	Control Flow Allocation (bytes)	Total Allocation (bytes)	SRAM Capacity (bytes)	Utilization
3_1	198,112	100,855	298,967	4,194,304	7.13%
4_1	117,888	53,676	171,564	4,194,304	4.09%
1_2	27,328	34,938	62,266	4,194,304	1.48%
2_2	164,864	82,991	247,855	4,194,304	5.91%
3_2	329,088	235,744	564,832	4,194,304	13.47%
4_2	158,464	99,233	257,697	4,194,304	6.14%
All	995,744	607,437	1,603,181	25,165,824	6.37%